River Test

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Ferry travelling across a river(40 points)

1. (4 points) A ferry is crossing a river that is 15m wide at a velocity of $3\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $3\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
2. (4 points) A ferry is crossing a river that is 192m wide at a velocity of $24\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $2\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
3. (4 points) A ferry is crossing a river that is 64m wide at a velocity of $16\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $8\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
4. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $24\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $2\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
5. (4 points) A ferry is crossing a river that is 180m wide at a velocity of $18\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $7\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat
6. (4 points) A ferry is crossing a river that is 100m wide at a velocity of $20\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $1\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river

(c) The displacement of the boat

(d) The vertical displacement of the boat

7. (4 points) A ferry is crossing a river that is 24m wide at a velocity of $4\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $4\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat

8. (4 points) A ferry is crossing a river that is 1m wide at a velocity of $1\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $1\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat

9. (4 points) A ferry is crossing a river that is 7m wide at a velocity of $1\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $9\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat

10. (4 points) A ferry is crossing a river that is 18m wide at a velocity of $18\,\mathrm{ms}^{-2}$ right relative to the water. However, the water is travelling at $4\,\mathrm{ms}^{-2}$ up relative to the ground. Calculate:
   (a) The resultant velocity of the ferry relative to the ground
   (b) The time taken to cross the river
   (c) The displacement of the boat
   (d) The vertical displacement of the boat